

# Seong-Jun Kang, Ph.D.

Experimental and computational biomedical scientist: human skin biology, single-cell and spatial profiling, machine-learning phenotyping

Postdoctoral Associate applicant · MIT Jameel Clinic · Yilmaz Lab (Koch Institute) and Barzilay Group (CSAIL)

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Available August 2026

## RESEARCH PROFILE

I run single-cell and spatial measurements on human tissue and build the models that read them. Five of my papers are on human skin, including atopic dermatitis, psoriasis, mycosis-fungoides-like inflammatory disease, and melanoma arising from a congenital melanocytic nevus, where I ran the GeoMx spatial workflow end to end and stained the S100B melanocyte marker by hand. In one immune-disease project I went from phenotyping 251,055 memory CD4<sup>+</sup> T cells to a target and then to a selective inhibitor that lowered the pathogenic fraction. My current work is deep-learning multimodal integration of imaging, transcriptomic and clinical data.

## POSITION-RELEVANT EXPERTISE

- **Human skin and pigmented tissue.** Five studies on atopic dermatitis, psoriasis, mycosis-fungoides-like inflammatory disease, and melanoma arising from congenital melanocytic nevus. Skin dissociation, IHC/IF on frozen and FFPE tissue, spatial profiling, and single-cell analysis.
- **Phenotyping and target discovery.** Disease-associated cell-state programs defined by **consensus NMF** from high-dimensional single-cell data, isoform-resolved analysis written from scratch, replication in an independent atlas, and then a causal and pharmacological test of the state.
- **Spatial and single-cell measurement.** **GeoMx DSP** end to end, **Visium**, **MACSima** cyclic multiplexed immunofluorescence, scRNA-seq, scTCR-seq, snATAC-seq, quantitative imaging, and multimodal integration across those layers.
- **Machine learning.** Deep-learning integration of imaging, transcriptomic and clinical data; a vision-language model adapted by LoRA so a pathology tile returns a structured reading, held out at the case level; **PyTorch**, scikit-learn, gradient boosting, calibrated classification, and evaluation held out at donor or case level.
- **Study design that a model can learn from.** Donor pooling with SNP demultiplexing so donor and run stay separable, multi-site QC across more than 700 donors, automated library preparation on the **10x Chromium Connect**, and plate, batch, donor and imaging round kept out of the label.
- **Grant writing.** About \$1.3 million across five industry proposals, lead writer on four, always under the institutional PI and not as PI. K99/R00 and LSRF are open to me, and I would apply within the first two years.

## EDUCATION

### DEGREES

|                           |                                                                                                                                                                                                                                         |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Ph.D.</b><br>2014–2023 | <b>Seoul National University</b> , Seoul, Republic of Korea<br>Department of Biomedical Sciences<br><i>Thesis: Exploration of the Role of Regulatory T cells in Various Pathogenic Contexts</i><br>Advisor: Chung-Gyu Park, M.D., Ph.D. |
| <b>M.S.</b><br>2012–2014  | <b>Seoul National University</b> , Seoul, Republic of Korea<br>Department of Biomedical Sciences<br><i>Thesis: Enhanced LPS-Induced Monocyte Activation by Recombinant Human Soluble CD14</i><br>Advisor: Chung-Gyu Park, M.D., Ph.D.   |
| <b>B.S.</b><br>2008–2012  | <b>Sungkyunkwan University</b> , Suwon, Republic of Korea<br>Department of Genetic Engineering                                                                                                                                          |

## RESEARCH & PROFESSIONAL EXPERIENCE

|                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Researcher<br>(postdoctoral)<br>2025–present | Yonsei University Wonju College of Medicine · Dept. of Physiology / Organelle Medicine<br>Research Center, Wonju, Republic of Korea<br>Deep-learning multimodal-integration models fusing imaging, transcriptomic, and clinical data into single predictive models.                                                                                                                                                                                                                                                                                                                                                            |
| Associate Director<br>2023–2025              | PB Immune Therapeutics Inc. · Dept. of Basic Research, Seoul, Republic of Korea<br>Led a humanized anti-CD40 antibody program: candidate ranking by <b>surface plasmon resonance (Biacore)</b> and ELISA, receptor-blocking confirmation, human and non-human-primate cross-reactivity, in vivo efficacy in primates, and the regulatory documentation. Placed and oversaw CRO and core-facility work, from study design and assay conditions through data delivery and go/no-go calls. Built the in-house single-cell and spatial analysis pipeline. Ran cynomolgus and marmoset EAE studies for multiple sclerosis programs. |
| Manager<br>2021–2023                         | PB Immune Therapeutics Inc. · Dept. of Basic Research, Seoul, Republic of Korea<br>Wrote the two KDDF proposals that funded the anti-CD40 program and ran the lead-selection work under them.                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Research Student<br>2020–2023                | SCAID, Single Cell Atlas of Immune Diseases, Genomic Medicine Institute, Seoul National University, Seoul, Republic of Korea<br>Contributed the founding proposal; executed within the funded program (700+ donors × 8 organs): milestone tracking, multi-site QC, multi-PI coordination.                                                                                                                                                                                                                                                                                                                                      |
| Researcher<br>2019–2021                      | Xenotransplantation Research Center, Seoul National University, Seoul, Republic of Korea<br>Rhesus macaque tissue processing and longitudinal immune monitoring across transplantation studies.                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Teaching Assistant<br>2014–2016              | Seoul National University College of Medicine · Dept. of Microbiology and Immunology, Seoul, Republic of Korea                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

## SELECTED RESEARCH OUTPUTS

- 1 Youngkyoung Lim\*, **Seong-Jun Kang\***, Beom Keun Cho, Hyun Je Kim, ... Chong Hyun Won, Chung-Gyu Park. "Intra-tumoral heterogeneity and immune escape of melanoma arising from congenital melanocytic nevus revealed by spatial gene expression profiling." *JEADV* 2022. [Download PDF](#)  
**Pigmented tissue measured in situ.** Ran GeoMx DSP end to end, including the S100B melanocyte-marker immunofluorescence stained by hand, fluorescence-guided ROI selection and instrument operation. The lesion was segmented by invasion depth from 300 to 4,200 µm Breslow, 16 whole-transcriptome ROIs; HLA class I and II were lost in a graded way with depth and B2M loss was confirmed by immunohistochemistry.
- 2 Woogil Song\*, **Seong-Jun Kang\***, Taehee Stella Kim\*, Seyoung Jung, Sung Hee Kim, Su Chan Park, Sung Hae Chang, Eun Young Lee, Tae-Gyun Kim, Hyun Je Kim, Jeong Seok Lee. "Conserved PDE4D short-isoform bias defines pathogenic Th1.17 state across human inflammatory diseases." *Nature Immunology* 2026. in revision  
**Phenotype, target, and drug in one project.** Custom UMI-aware, BAM-level code resolved PDE4D short- and long-isoform identity from 5' reads across 251,055 cells. The pathogenic Th1.17 program replicated across 702 samples and 23 immune diseases. Short-isoform overexpression raised the IFN-γ<sup>+</sup>IL-17A<sup>+</sup> fraction, the long isoform left it flat, and the PDE4D-selective inhibitor zatolmilast lowered it.
- 3 **Seong-Jun Kang\***, Jeong-Ryeol Gong\*, Seon-Pil Jin, Jin-Mi Oh, Hyunjin Jin, Yuji Lee, Yewon Moon, Dongjun Kim, Hyo Jeong Nam, Hyun Seung Choi, Sanha Hwang, Yun Jung Huh, Kyung Yeon Han, Jihwan Moon, Jongsuk Chung, Woong-Yang Park, Chung-Gyu Park, Hyun Je Kim, Jeong Eun Kim. "Deciphering Dysfunctional Regulatory T Cells in Atopic Dermatitis." *Allergy* 2025. [Download PDF](#)  
**Human skin immunology at single-cell resolution.** Single-cell and functional dissection of skin-homing CLA<sup>+</sup> regulatory T cells in atopic dermatitis, with OX40 up-regulated alongside impaired suppressive function.

- 4 Jung Ho Lee\*, Sung Ha Lim\*, Hyungdon Kook, **Seong-Jun Kang**, Geon Lee, Brian Hyohyoung Lee, Hyunsung Nam, YongJun Kim, Christine Suh-Yun Joh, Soyoung Jeong, Dongryeol Shin, Hyun Je Kim, Jiyoung Ahn. "Mycosis Fungoides-like Atopic Dermatitis Represents a Th22-dominant Inflammatory Endotype." *Allergy* 2026.  
**Telling a reactive state from a malignant one.** Cross-dataset scVI integration with public cohorts, InferCNV genomic-identity inference and scTCR clonality across 248,881 cells, showing an oligoclonal Th22 inflammatory endotype rather than a malignancy.
- 5 Brian Hyohyoung Lee\*, Yoon Ji Bang\*, Sung Ha Lim\*, **Seong-Jun Kang**, Sung Hee Kim, Seunghee Kim-Schulze, Chung-Gyu Park, Hyun Je Kim, Tae-Gyun Kim. "High-dimensional profiling of regulatory T cells in psoriasis reveals an impaired skin-trafficking property." *eBioMedicine* 2024. [Download PDF](#)  
**High-dimensional profiling of inflamed skin.** Regulatory T cells in psoriasis profiled at high parameter, showing an impaired skin-trafficking phenotype.

## TECHNICAL EXPERTISE

|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Machine-learning phenotyping         | Current work at Yonsei is <b>deep-learning multimodal integration</b> , fusing imaging, transcriptomic and clinical data into single predictive models. Cell-state programs defined by <b>consensus NMF</b> ( $k = 14 \rightarrow 12$ reproducible programs across 251,055 cells), then a condition classifier on subset composition reported with held-out AUC and 95% confidence intervals (arthritis 0.780, healthy control 0.771, psoriasis 0.806). <b>PyTorch</b> , scikit-learn, XGBoost/LightGBM/CatBoost. Quantitative image analysis on IHC/IF, frozen and FFPE.                                                                                                                                                                                            |
| Skin and pigmented tissue            | Five human-skin papers: atopic dermatitis ( <i>Allergy</i> 2025, co-first), psoriasis ( <i>eBioMedicine</i> 2024), mycosis-fungoides-like AD ( <i>Allergy</i> 2026), and two on melanoma arising from congenital melanocytic nevus ( <i>JEADV</i> 2022 co-first, 2024). <b>GeoMx DSP</b> run end to end on the nevus melanoma, including S100B melanocyte-marker immunofluorescence stained by hand, depth-segmented sampling, and 16 whole-transcriptome ROIs. Skin tissue dissociation, frozen and FFPE IHC/IF.                                                                                                                                                                                                                                                    |
| Single-cell / spatial / multi-omic   | scRNA-seq, scTCR-seq, snRNA/snATAC; 10x 5'/3' library generation and <b>10x Chromium Connect</b> automated preparation. <b>Seurat</b> , <b>Scanpy</b> , <b>Squidpy</b> , <b>Monocle3</b> , <b>CellChat/NicheNet</b> ; scVI/scANVI, batch integration, <b>consensus NMF</b> ; Milo KNN-graph differential abundance, GLIPH, InferCNV. Custom UMI-aware, BAM-level isoform analysis code (Python). Spatial: <b>GeoMx DSP</b> run end to end, <b>Visium</b> , <b>MACSima</b> cyclic multiplexed immunofluorescence.                                                                                                                                                                                                                                                     |
| Vision-language and retrieval models | <b>Finetuning.</b> LoRA adaptation of an open vision-language model ( <b>Gemma 4 12B</b> , mlx-vlm) so that an amyloid pathology tile returns a structured JSON reading, with the schema enforced during decoding and the split held out at the case level. Against the same model zero-shot on 800 held-out tiles, cored-plaque F1 went from 0.12 to 0.44 and cerebral amyloid angiopathy from 0.00 to 0.69. <b>Retrieval.</b> A retrieval-augmented question-answering service written end to end: corpus chunking, BM25 and dense-vector retrieval fused by reciprocal rank, query expansion, and streamed answers that carry their sources. <b>FastAPI</b> , containerized, on <b>Google Cloud Run</b> ; it answers questions about this CV at kangseongjun.com. |
| Computational                        | Python: scikit-learn, XGBoost/LightGBM/CatBoost, <b>PyTorch</b> ; AUC-validated classifiers held out on independent cohorts; multimodal deep-learning integration (imaging + transcriptomic + clinical). Linux servers · R/RStudio · Git-based reproducible analysis.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Cell, tissue & immunological assays  | Primary human T-cell isolation, expansion and functional assays; monocyte and dendritic-cell differentiation; mesenchymal stem cell culture. Enzymatic tissue digestion, Percoll enrichment, cell and nucleus isolation. High-dimensional flow cytometry (panels up to 24 markers); multiplex panel design; 10-color 6-way FACS. <b>Olink</b> targeted plasma proteomics (immune & neurology panels).                                                                                                                                                                                                                                                                                                                                                                |

## Microscopy & animal models

Spinning-disk confocal (Nikon CSU-W1 SORA), point-scanning confocal (Leica SP8), two-photon; IHC/IF (frozen & FFPE) with quantitative image analysis. Non-human primates across three species (rhesus, cynomolgus, marmoset), including EAE and islet-transplantation studies; mouse immune profiling.

## Program & external execution

CRO and core-facility work from study design and placement through data delivery and go/no-go calls; assay conditions specified to the vendor; regulatory documentation for a therapeutic antibody program; funder milestones and reporting; multi-site sample QC across a 700-donor collection.

## PUBLICATIONS

18

TOTAL PAPERS

17

PEER-REVIEWED

1

IN REVISION

6

FIRST / CO-FIRST

 [Google Scholar · Full Profile, Citations & h-index ↗](#)

### PUBLICATIONS · FIRST / CO-FIRST AUTHOR

2026

Woogil Song\*, **Seong-Jun Kang\***, Taehee Stella Kim\*, Seyoung Jung, Sung Hee Kim, Su Chan Park, Sung Hae Chang, Eun Young Lee, Tae-Gyun Kim, Hyun Je Kim, Jeong Seok Lee. "Conserved PDE4D short-isoform bias defines pathogenic Th1.17 state across human inflammatory diseases." *Nature Immunology*. [in revision](#)

2025

**Seong-Jun Kang\***, Yong-Hee Kim\*, Thuy Nguyen-Phuong\*, Yijoon Kim, Jin-Mi Oh, Jae-chun Go, DaeSik Kim, Chung-Gyu Park, Hyunsu Lee, Hyun Je Kim. "Immune cell-enriched single-cell RNA sequencing unveils the interplay between infiltrated CD8<sup>+</sup> T resident memory cells and choroid plexus epithelial cells in Alzheimer's disease." *Journal of Neuroimmunology*. [Download PDF](#)

Co-first author. Built the brain-to-single-cell path from whole-brain harvest through enzymatic digestion, Percoll enrichment and FACS to 10x; identified exhausted CD8<sup>+</sup> tissue-resident memory T cells engaging choroid-plexus epithelium at the blood-CSF barrier.

2025

**Seong-Jun Kang\***, Jeong-Ryeol Gong\*, Seon-Pil Jin, Jin-Mi Oh, Hyunjin Jin, Yuji Lee, Yewon Moon, Dongjun Kim, Hyo Jeong Nam, Hyun Seung Choi, Sanha Hwang, Yun Jung Huh, Kyung Yeon Han, Jihwan Moon, Jongsuk Chung, Woong-Yang Park, Chung-Gyu Park, Hyun Je Kim, Jeong Eun Kim. "Deciphering Dysfunctional Regulatory T Cells in Atopic Dermatitis." *Allergy*. [Download PDF](#)

2024

Jong-Min Kim\*, **Seong-Jun Kang\***, So-Hee Hong\*, Hyunwoo Chung, Jun-Seop Shin, Byoung-Hoon Min, Hyun Je Kim, Jongwon Ha, Chung-Gyu Park. "Long-term control of diabetes by tofacitinib-based immunosuppressive regimen after allo-islet transplantation in diabetic rhesus monkeys that rejected previously transplanted porcine islets." *Xenotransplantation*. [Download PDF](#)

Co-first author. Long-term diabetes control with a tofacitinib-based regimen after allo-islet transplantation in rhesus macaques, with longitudinal immune monitoring.

2022

Youngkyoung Lim\*, **Seong-Jun Kang\***, Beom Keun Cho, Hyun Je Kim, ... Chong Hyun Won, Chung-Gyu Park. "Intra-tumoral heterogeneity and immune escape of melanoma arising from congenital melanocytic nevus revealed by spatial gene expression profiling." *JEADV*. [Download PDF](#)

2019

Jong-Min Kim\*, Jun-Seop Shin\*, Byoung-Hoon Min\*, **Seong-Jun Kang\***, Il-Hee Yoon, Hyunwoo Chung, Jiyeon Kim, Eung-Soo Hwang, Jongwon Ha, Chung-Gyu Park. "JAK3 inhibitor-based immunosuppression in allogeneic islet transplantation in cynomolgus monkeys." *Islets*. [Download PDF](#)

Co-first author, one of four authors contributing equally. JAK3-inhibitor-based immunosuppression in allogeneic islet transplantation in cynomolgus macaques.

## CONTRIBUTING AUTHOR

- 2026 Jung Ho Lee\*, Sung Ha Lim\*, Hyungdon Kook, **Seong-Jun Kang**, Geon Lee, Brian Hyohyoung Lee, Hyunsung Nam, YongJun Kim, Christine Suh-Yun Joh, Soyoung Jeong, Dongryeol Shin, Hyun Je Kim, Jiyoung Ahn. "Mycosis Fungoides-like Atopic Dermatitis Represents a Th22-dominant Inflammatory Endotype." *Allergy*.
- 2024 Brian Hyohyoung Lee\*, Yoon Ji Bang\*, Sung Ha Lim\*, **Seong-Jun Kang**, Sung Hee Kim, Seunghee Kim-Schulze, Chung-Gyu Park, Hyun Je Kim, Tae-Gyun Kim. "High-dimensional profiling of regulatory T cells in psoriasis reveals an impaired skin-trafficking property." *eBioMedicine*. [Download PDF](#)
- 2024 Youngkyoung Lim\*, Beom Keun Cho\*, **Seong-Jun Kang**, Soyoung Jeong, Hyun Je Kim, Jiyeon Baek, Ji Hwan Moon, Cheol Lee, Chan-Sik Park, Je-Ho Mun, Chong Hyun Won, Chung-Gyu Park. "Spatial transcriptomic analysis of tumour-immune cell interactions in melanoma arising from congenital melanocytic nevus." *JEADV*. [Download PDF](#)
- 2023 Sunyoung Jung\*, Sunho Lee, Hyun Je Kim, Sueon Kim, Ji Hwan Moon, Hyunwoo Chung, **Seong-Jun Kang**, Chung-Gyu Park. "Mesenchymal stem cell-derived extracellular vesicles subvert Th17 cells by destabilizing RORyt through posttranslational modification." *Experimental & Molecular Medicine*.
- 2022 Sangho Lee, Han-Teo Lee, Young Ah Kim, Il-Hwan Lee, **Seong-Jun Kang**, Kyeongpyo Sim, Chung-Gyu Park, Kyungho Choi, Hong-Duk Youn. "The optimized core peptide derived from CABIN1 efficiently inhibits calcineurin-mediated T-cell activation." *Experimental & Molecular Medicine*.
- 2021 So Hee Hong, Hyun Je Kim, **Seong-Jun Kang**, Chung-Gyu Park. "Novel Immunomodulatory Approaches for Porcine Islet Xenotransplantation." *Current Diabetes Reports*. [Download PDF](#)
- 2020 Sunho Lee, Sueon Kim, Hyunwoo Chung, Ji Hwan Moon, **Seong-Jun Kang**, Chung-Gyu Park. "Mesenchymal stem cell-derived exosomes suppress proliferation of T cells by inducing cell cycle arrest through p27kip1/Cdk2 signaling." *Immunology Letters*.
- 2019 Hyun Je Kim, Ji Hwan Moon, Hyunwoo Chung, Jun-Seop Shin, Bonggi Kim, Jong Min Kim, Jung-Sik Kim, Il-Hee Yoon, Byoung-Hoon Min, **Seong-Jun Kang**, Yong-Hee Kim, Kyuri Jo, Joungmin Choi, Heejoon Chae, Won Woo Lee, Sun Kim, Chung-Gyu Park. "Bioinformatic analysis of peripheral blood RNA-sequencing sensitively detects the cause of late graft loss following overt hyperglycemia in pig-to-non-human primate islet xenotransplantation." *Scientific Reports*.
- 2018 Byoung-Hoon Min, Jun-Seop Shin, Jong-Min Kim, **Seong-Jun Kang**, Hyun-Je Kim, Il-Hee Yoon, Su-Kyoung Park, Ji-won Choi, Min-Suk Lee, Chung-Gyu Park. "Delayed revascularization of islets after transplantation by IL-6 blockade in pig-to-non-human primate islet xenotransplantation model." *Xenotransplantation*.
- 2018 Jun-Seop Shin, Jong-Min Kim, Byoung-Hoon Min, Il Hee Yoon, Hyun Je Kim, Jung-Sik Kim, Yong-Hee Kim, **Seong-Jun Kang**, Jiyeon Kim, Hee-Jung Kang, Dong-Gyun Lim, Eung-Soo Hwang, Jongwon Ha, Sang-Joon Kim, Wan Beom Park, Chung-Gyu Park. "Pre-clinical results in pig-to-non-human primate islet xenotransplantation using anti-CD40 antibody (2C10R4)-based immunosuppression." *Xenotransplantation*.
- 2018 Jong-Min Kim, Jun-Seop Shin, Byoung-Hoon Min, Il Hee Yoon, **Seong-Jun Kang**, Won-Young Jeong, Sang-Joon Kim, Chung-Gyu Park. "Pre-Clinical Results of Islet Allo-Transplantation Using JAK Inhibitor as Replacement for Tacrolimus in Cynomolgus Monkeys." *Transplantation*.
- 2018 Jung-Sik Kim, Hyunwoo Chung, Nari Byun, **Seong-Jun Kang**, Sunho Lee, Jun-Seop Shin, Chung-Gyu Park. "Construction of EMSC-islet co-localizing composites for xenogeneic porcine islet transplantation." *Biochemical and Biophysical Research Communications*.

2026

Kangsan Roh\*, Sabyasachi Das\*, **Seong-Jun Kang**, Olga Barreiro, Rodrigo Javier Gonzalez, Timothy P. Padera, Shraddha Shirguppe, Ahlim Lee, Youngju Song, Pazhanichamy Kalailingam, Claire E. Fong, Aarushi Gandhi, Meehyun Kim, Diana Tambala, Adam Vernon Crain, Casey A. Maguire, Ulrich H. Von Andrian, Christian L. Lino Cardenas, Sekeun Kim, Benjamin P. Kleinstiver, Mark E. Lindsay, Patricia L. Musolino, Rajeev Malhotra. "In-vivo base editing of ACTA2 R179H restores lymphatic function and immune cell homeostasis in a murine model of Multisystemic Smooth Muscle Dysfunction Syndrome (MSMDS)." In preparation

Co-author (immunology). Ran the Olink plasma proteomics across 12 patients and 6 controls and the flow-cytometry immunophenotyping of lymph-node leukocytes.

SELECTED GRANT WRITING & PROGRAM FUNDING

INDUSTRY · PB IMMUNE THERAPEUTICS (MANAGER – ASSOCIATE DIRECTOR)

| FUNDING AGENCY / SPONSOR            | PROJECT                                                                                       | PERIOD    | BUDGET (USD) | ROLE / CONTRIBUTION                                 |
|-------------------------------------|-----------------------------------------------------------------------------------------------|-----------|--------------|-----------------------------------------------------|
| Korea Drug Development Fund (KDDF)  | Anti-CD40 antibody lead selection and optimization (multiple sclerosis)                       | 2021–2023 | ~\$571,000   | Lead proposal writer (>90% of proposal & execution) |
| Korea Drug Development Fund (KDDF)  | Anti-CD40 lead compound (PB101): efficacy and pharmacodynamic biomarkers (pemphigus vulgaris) | 2022–2024 | ~\$321,000   | Lead proposal writer (>90% of proposal & execution) |
| Ministry of SMEs and Startups (MSS) | Scale-up Funding for Tech Commercialization                                                   | 2022–2023 | ~\$222,000   | Lead proposal writer; managed >90% of execution     |
| Ministry of SMEs and Startups (MSS) | Early-stage startup R&D: anti-CD40 biologic (PB-40N) to prevent anti-drug antibodies          | 2021–2022 | ~\$74,000    | Lead proposal writer & core researcher              |
| NAVER Corporation                   | Engineered regulatory T cells for immune-mediated diseases (PB20X_Treg)                       | 2023–2026 | ~\$128,000   | Proposal & study design (~50%)                      |

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| FUNDING AGENCY / SPONSOR                                         | PROJECT                                                                                         | PERIOD          | BUDGET (USD) | ROLE / CONTRIBUTION                                       |
|------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|-----------------|--------------|-----------------------------------------------------------|
| Ministry of Science and ICT (MSIT)                               | Multi-layered immune-disease single-cell map for molecular target discovery (SCAID, A9-22-1-05) | 2022–2028 (3+3) | ~\$1.6M      | Participated in proposal writing (>30%)                   |
| Ministry of Education (MOE) & Ministry of Science and ICT (MSIT) | Laboratory-Specialized Tech-Transfer & R&BD Grant                                               | 2020–2021       | ~\$40,000    | Grant writing, experimental design & technical validation |

All grants were held by the institutional Principal Investigator (Ph.D. advisor at SNU; company CEO/CSO at PB Immune Therapeutics); the listed roles reflect the applicant's contribution to proposal writing and project execution, **not as PI**. Budgets converted at ≈ ₩1,400 / USD.

AWARDS, PATENTS & SCHOLARSHIPS

PATENTS

|      |                                                                                                                              |
|------|------------------------------------------------------------------------------------------------------------------------------|
| 2024 | Antagonistic Anti-CD40 Humanized Antibody. Republic of Korea, Appl. No. 10-2024-0085972; registered 2024-07-01. Co-inventor. |
| 2019 | Immunosuppression Composition Comprising JAK Inhibitor. Republic of Korea, Appl. No. 10-2014-3490000; registered 2019-08-20. |

## AWARDS

- |      |                                                                                                                         |
|------|-------------------------------------------------------------------------------------------------------------------------|
| 2024 | Grand Prize, 2024 GPTers AI Hackathon (Project GAIA Team)                                                               |
| 2023 | Grand Prize, SBA Smart Workathon (Work Process Innovation)                                                              |
| 2017 | Best Poster Award, American Transplant Congress, Chicago · Best Presentation Award, Korean Association of Immunologists |

## SCHOLARSHIPS

- |           |                                                                                                                                                         |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2014–2020 | NRF Research Scholarship for Ph.D. Candidates · Dean's Lecture & Research Support Scholarship, SNU College of Medicine · BK21 Plus Research Scholarship |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------|

## ■ SELECTED PRESENTATIONS

- |        |                                                                                                                                                                                                                       |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Poster | KAI Annual Meeting, Incheon (2022) · Cytokine, Hawaii (2022) · American Transplant Congress, Chicago (2017) · The Transplantation Society, Hong Kong (2016) · IPITA/IXA/CTS, Melbourne (2015) · TREG, Shanghai (2012) |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## ■ REFERENCES

### Chung–Gyu Park, M.D., Ph.D.

Professor, Microbiology & Immunology / Biomedical Sciences, SNU College of Medicine; CEO/CSO, PB Immune Therapeutics · chgpark@snu.ac.kr

*Doctoral advisor and industry supervisor; directed the anti-CD40 antibody program.*

### Seung–Kuy Cha, Ph.D.

Professor, Dept. of Physiology & Global Medical Science, Yonsei University Wonju College of Medicine · skcha@yonsei.ac.kr

*Most recent research supervisor (2025–2026).*

### Hyun Je Kim, M.D., Ph.D.

Assistant Professor, Microbiology & Immunology / Biomedical Sciences; Interdisciplinary Program in Artificial Intelligence (IPAI), SNU · tte9801@snu.ac.kr

*Co-corresponding on the single-cell studies; supervised the computational work.*

### Jeong Seok Lee, M.D., Ph.D.

Associate Professor, Graduate School of Medical Science & Engineering, KAIST, Daejeon · chemami@kaist.ac.kr

*Corresponding author, PDE4D / Nature Immunology study.*

|                                                                                                                                                            |                                                  |                     |
|------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|---------------------|
| Seong–Jun Kang · Curriculum Vitae Available from August 2026 · based in Boston, MA · currently authorized to work in the United States · MIT Jameel Clinic | without employer sponsorship to begin employment | Updated August 2026 |
|------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|---------------------|